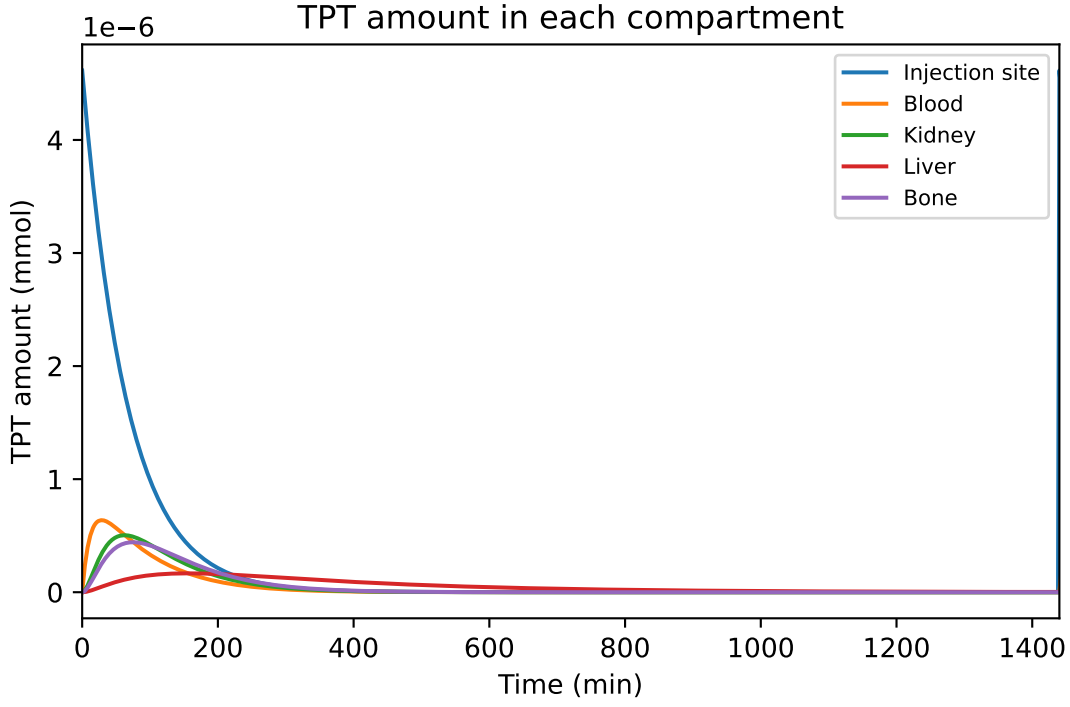
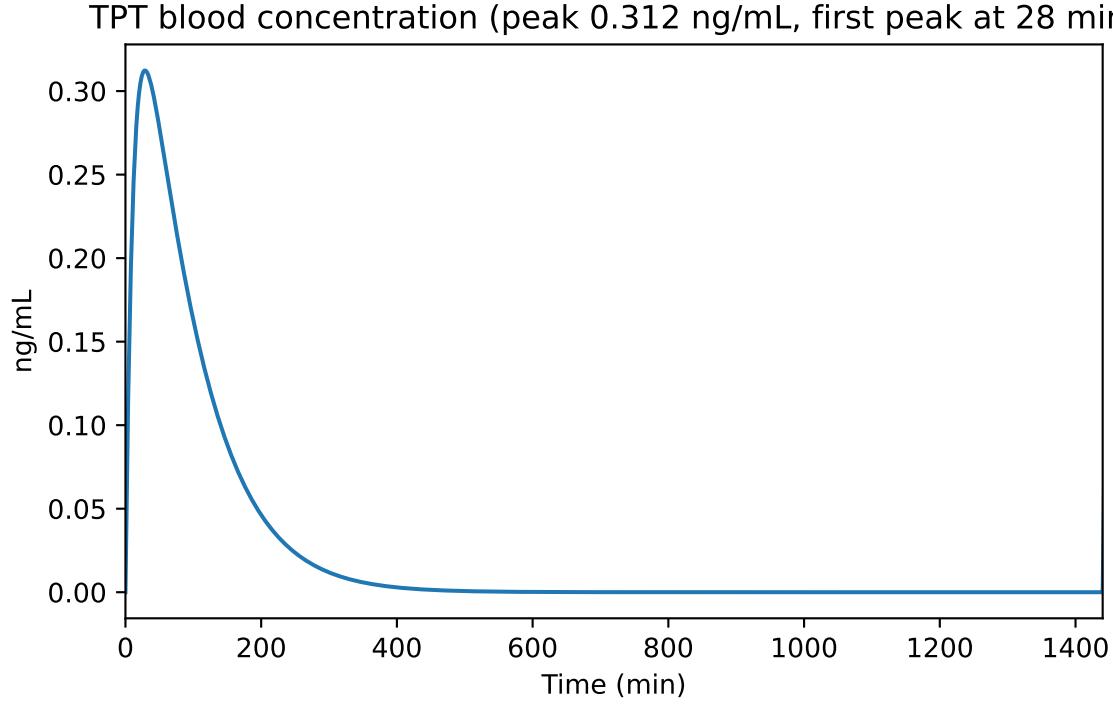
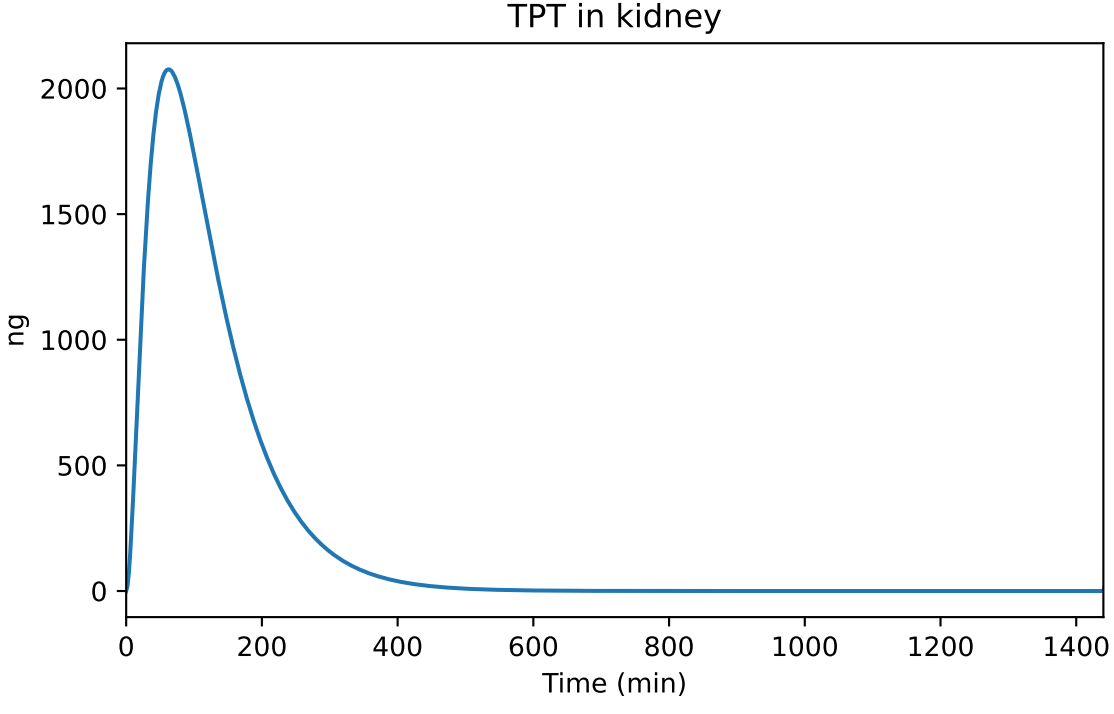
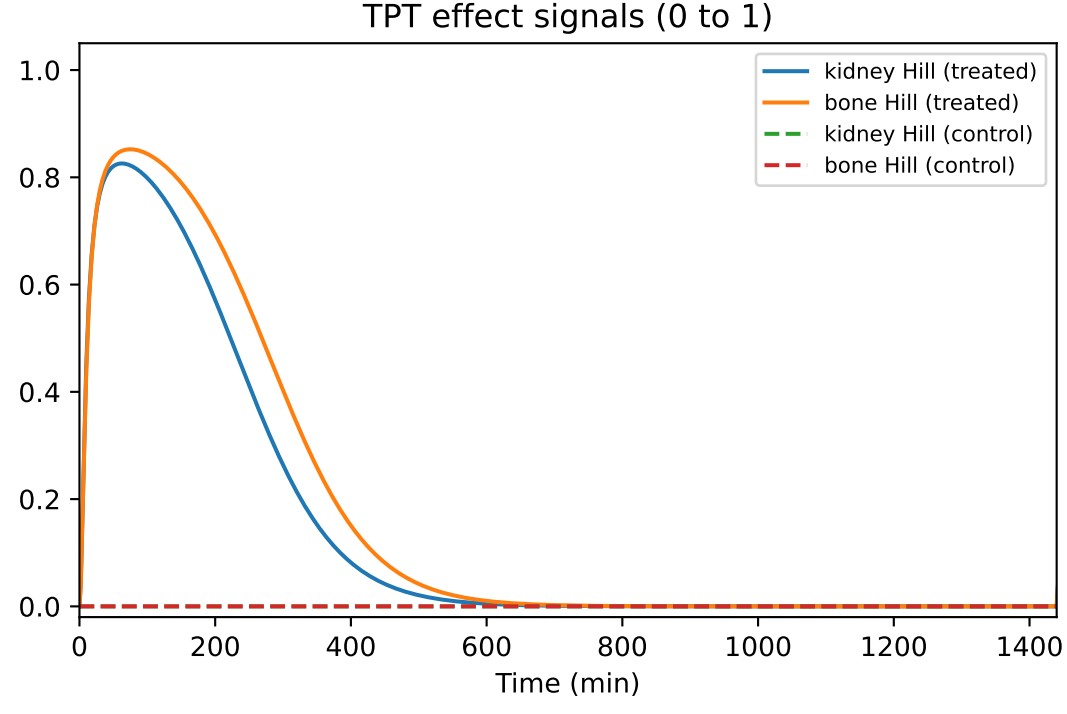
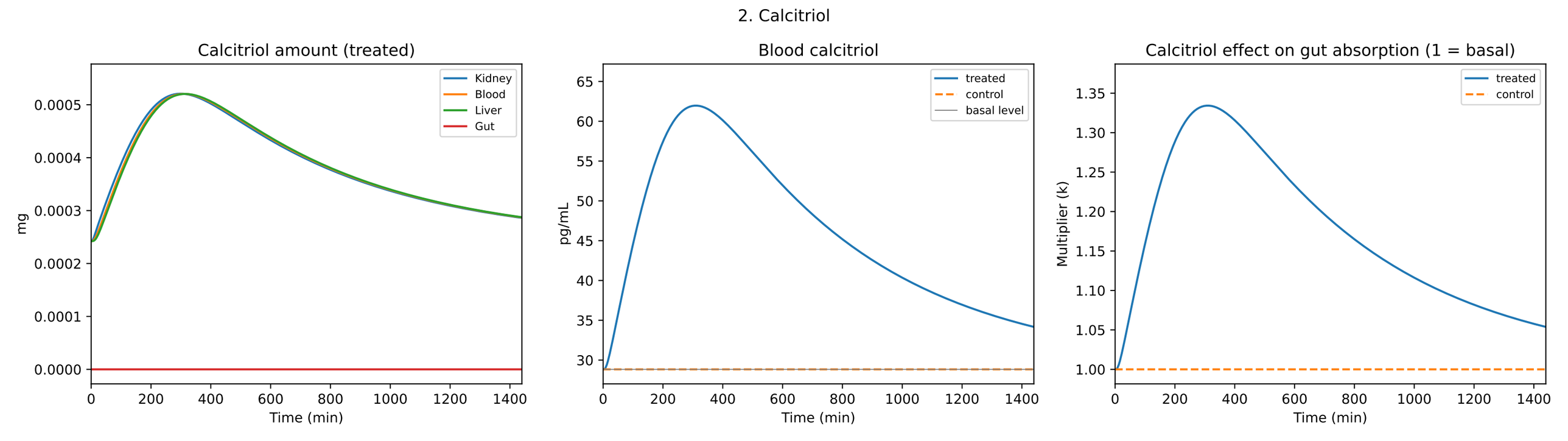
1. TPT pharmacokinetics

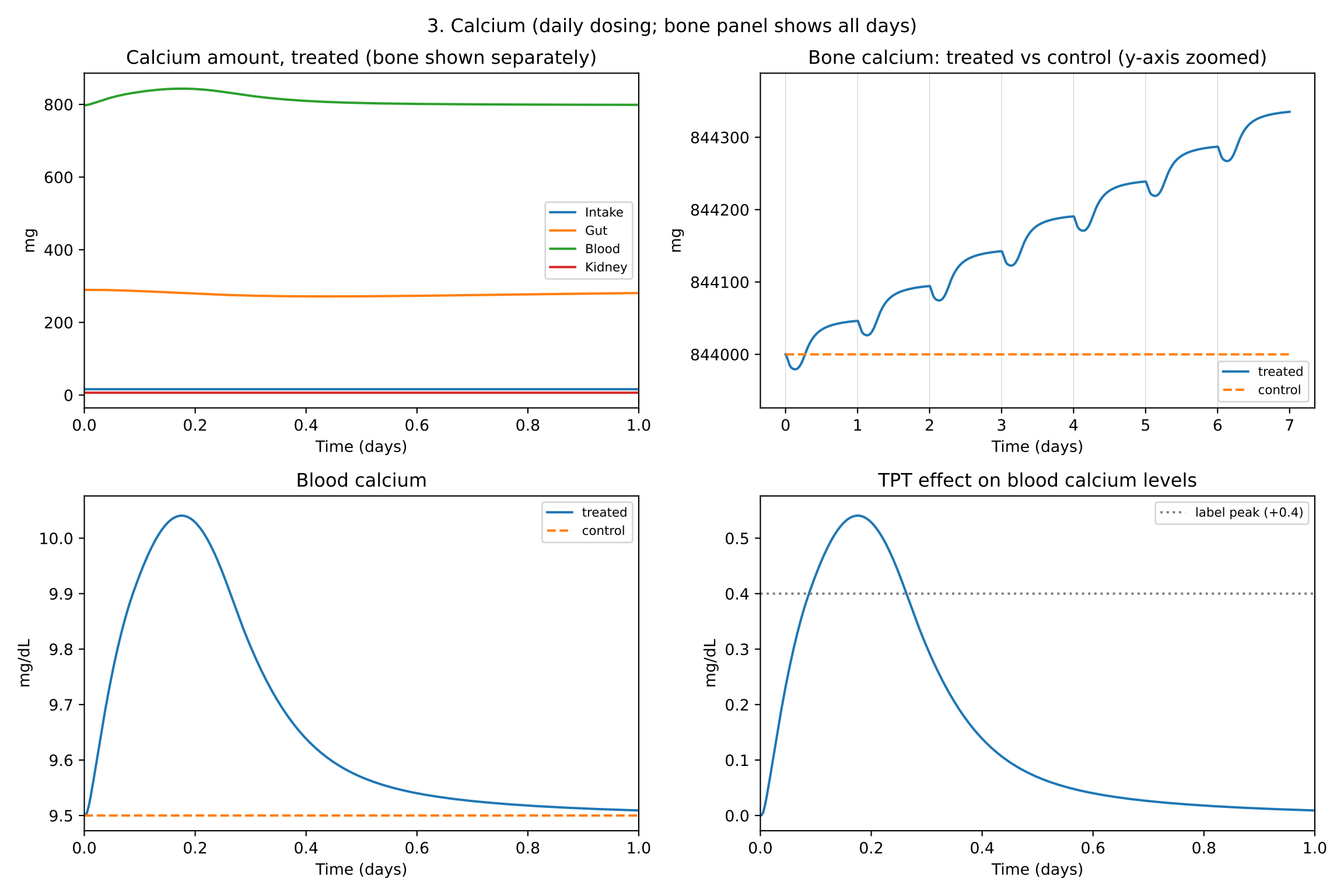


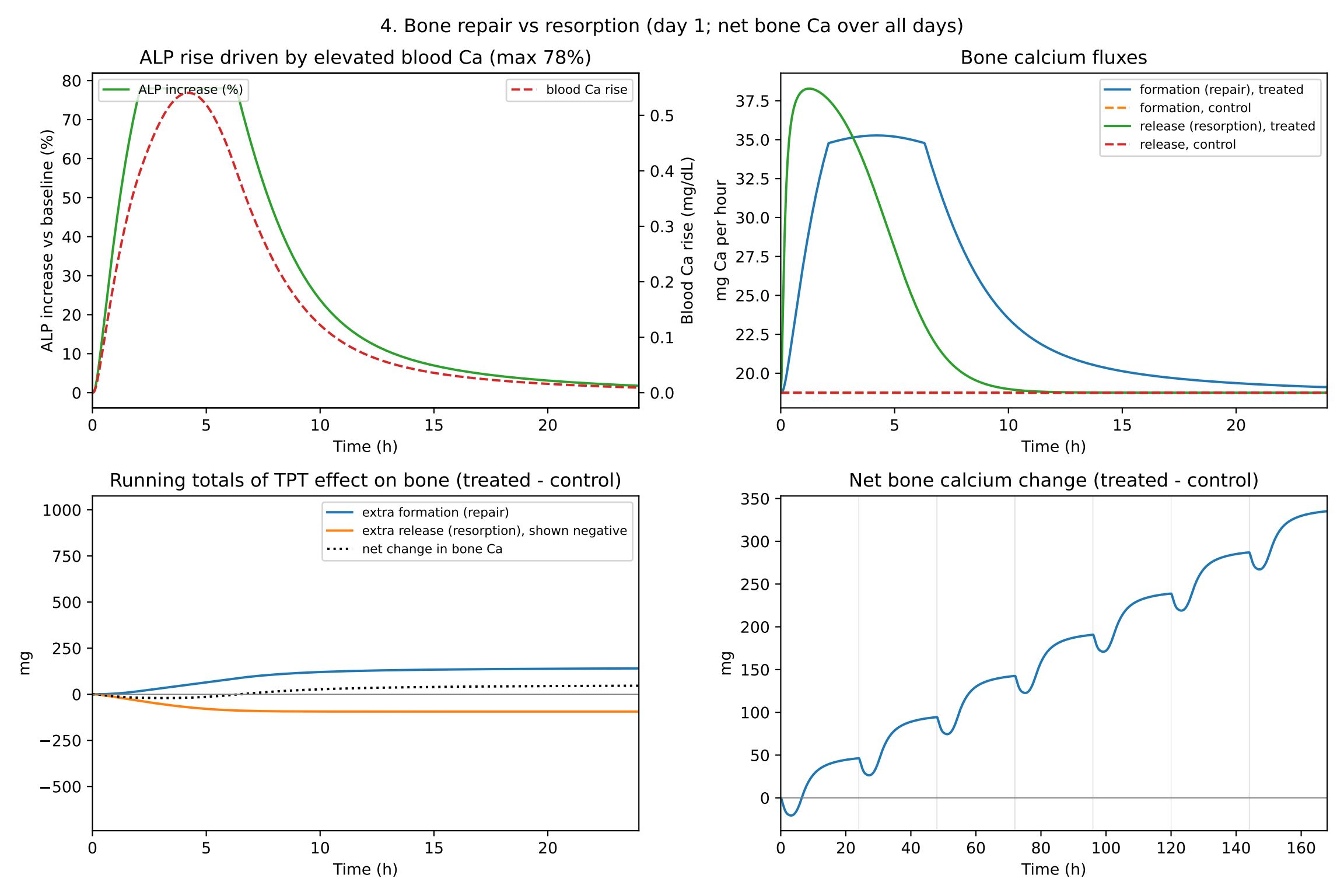







2. Calcitriol
Calcitriol amount (treated)
Kidney
Blood
Liver
Gut
mg
0.0005
0.0004
0.0003
0.0002
0.0001
0.0000
0
200
400
600
800
1000
1200
1400
Time (min)
Blood calcitriol
treated
control
basal level
pg/mL
65
60
55
50
45
40
35
30
0
200
400
600
800
1000
1200
1400
Time (min)
Calcitriol effect on gut absorption (1 = basal)
treated
control
Multiplier (k)
1.35
1.30
1.25
1.20
1.15
1.10
1.05
1.00
0
200
400
600
800
1000
1200
1400
Time (min)


3. Calcium (daily dosing; bone panel shows all days)
Calcium amount, treated (bone shown separately)
mg
Time (days)
Intake
Gut
Blood
Kidney
Bone calcium: treated vs control (y-axis zoomed)
mg
Time (days)
treated
control
Blood calcium
mg/dL
Time (days)
treated
control
TPT effect on blood calcium levels
mg/dL
Time (days)
label peak (+0.4)


4. Bone repair vs resorption (day 1; net bone Ca over all days)
ALP rise driven by elevated blood Ca (max 78%)
ALP increase (%)
blood Ca rise
ALP increase vs baseline (%)
Blood Ca rise (mg/dL)
Time (h)
Bone calcium fluxes
formation (repair), treated
formation, control
release (resorption), treated
release, control
mg Ca per hour
Time (h)
Running totals of TPT effect on bone (treated - control)
extra formation (repair)
extra release (resorption), shown negative
net change in bone Ca
mg
Time (h)
Net bone calcium change (treated - control)
mg
Time (h)